Headwater

Duty Engineer · Ashfell Dam

15 DAYS · 52 STOPS · GRADE 12 · HEADWATER

THE OPENING CARD

Ashfell is 88% full, and nine days of rain start on Thursday. This was the driest summer in nine years, so no one here wants to let water go. You are the duty engineer, and each morning you say how much goes out. You say it from gauge readings hours old and a chart of the lake bed drawn in 2003. In fifteen days you hand over the Ashfell release rules, and four villages live on the river below the wall.

DAY 1 OF 15

88%

BEFORE THE DAY — GREET

Five floors, one shift, and the spillway is running

Put a face to each desk before the wet season: five floors, one stair between them, and the people who read the gauges work three floors above the people who wind the gates.

PLAN CARD 137W

Monday. June Sato, the forecast liaison, brought the five-day forecast down an hour ago. Rain over the high ground from Thursday. A lot of it. Dermot Halloran, the operations manager, has already said the word drawdown. He said it in a tone that means he does not want one. Tonight the first of the release rules goes on the board in the storage room. Every rule after it is measured against a lake-bed survey done in 2003 and never done again. So this fortnight is not run on readings. It is run on rates. A rate is how fast a number is changing. Today you find a rate three ways. Once between two gauge readings. Once from a fitted formula, straight from the definition. Once at a single instant, where the record holds no rate at all.

BRIEFING 21W

Nine days of rain forecast, and every number on this dam is about to be asked how fast it is moving.

WHAT YOU DECIDE 18W

Get an instantaneous rate out of readings, out of a formula, and find the instant that has none.

1.1 The slope between two readings

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · BALLPARK

SCENE 42W

Ayo Ekundayo, the catchment hydrologist, has the Ashfell gauge on the screen — 41 cubic metres a second at 6 this morning, 48 at 8. Halloran wants a number for how fast it is climbing, not another reading off the same trace.

GUIDE 72W

Five numbers are on offer and only three of them are ends of an interval. Two are arithmetic already done to the readings: a sum, and the hour one of them was taken at. Ask of each what it measures. A sum of two readings has thrown away the difference a slope is made of. Get that wrong and the report gives the river's size where it was asked for its speed.

ASK 14W

Estimate the average rate of change of the inflow between six and eight o'clock.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

a flow rate says how much water passes a point each second · the slope of a line through two points is the rise divided by the run

VERDICT 82W

An average rate of change is a slope, and the two readings are the two ends of it. It says nothing about what happened between them. The same pair comes from a river that climbed steadily. It also comes from one that sat still for 90 minutes and then jumped. Shrinking the interval is the only cure. That shrinking approaches a number: the limit of this slope as the two readings close together. That limit is the instantaneous rate the forecast wants.

TAKEAWAY 16W

An average rate of change is the slope of the line joining an interval's two endpoints.

1.2 What the slope becomes

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A PERSON · DERIVE

SCENE 43W

Bo Ferrand, the records & rating curves, has fitted last night's storage to $V(t) = 5t^2 + 40t$, in thousands of cubic metres, with t in hours. The nine o'clock call wants the rate at four hours, not an average across the night.

GUIDE 59W

Build it a line at a time from the definition — the difference quotient — rather than from a rule you already know. Each step is the expression the line above licenses, and every wrong branch is legal algebra. The nine o'clock call wants the rate at four hours, so finish by evaluating what you derived at that value.

ASK 15W

Work the difference quotient down to an instantaneous rate, and evaluate it at four hours.

BEHIND THE BUTTON 174W

Why the definition rather than the rule. The power rule gives the answer in one line and teaches nothing about why the answer is right. Working the difference quotient through is where the rule comes from. That means the change in V over the change in t, as that change goes to nothing. It is the only version that survives a function no rule covers.

What the algebra does. Expand $V(t + h)$, subtract $V(t)$, and every term without an h cancels. Divide by h and one factor of h comes out of what is left. The limit is then the terms with no h in them at all, which is the derivative — and the cancellation is the whole trick.

Why an average will not do. Storage over the night rose in a curve. So an average across it is a rate that was true at one instant somewhere in the middle. It is wrong now. The call is about what to do next, and that depends on the rate at four hours.

TAKES AS READ

an average rate of change is the change in a quantity over the change in time · a limit is the value an expression approaches as one of its inputs closes on a number · limits, one-sided limits, and where a limit fails to exist — taken as read · continuity, and what a jump costs you — taken as read

VERDICT 85W

The difference quotient is the slope of a line through two points on the curve, and its value depends on how far apart they are. Taking the limit closes that gap without ever setting it to 0. That is the whole manoeuvre. At $h = 0$ the quotient is $0/0$ and says nothing, while its limit is one number. What comes out is a rule, $10t + 40$, good at every hour of the night, and the four-hour figure is one substitution away from it.

TAKEAWAY 18W

> The derivative is the limit of the difference quotient, and taking the limit is what removes the interval.

1.3 The instant the gate moved

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · CHOICE

SCENE 40W

Nadine Baptiste, the downstream warning officer, has the tailrace trace across the moment gate 2 opened. Discharge sits at 60 cubic metres a second up to 2 o'clock and at 145 immediately after. She wants the trace read, not smoothed.

GUIDE 61W

The four answers disagree about one thing: whether a rate exists at that instant at all. Some assign it a value, large or zero, and some deny there is a value to assign. Decide that first, from the trace, before arguing about size. Sorting a discontinuity from a steep slope decides whether you need a better gauge or a different question.

ASK 9W

What is true of the discharge at two o'clock?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

a limit describes what a quantity approaches from one side, which need not be what it reaches · a derivative is itself a limit, taken across the instant rather than up to it · limits, one-sided limits, and where a limit fails to exist — taken as read

VERDICT 83W

Approaching 2 o'clock from the left, the discharge tends to 60. From the right it tends to 145. A derivative is the limit of a difference quotient taken across the instant. Here that quotient is a fixed jump divided by a shrinking interval, so it grows without bound. It never settles on a number. The trace has a good rate everywhere else. That is why asking for the rate of rise at two o'clock asks the record for something it does not hold.

> TAKEAWAY 24W
>
> A jump in a record is a place the calculus simply does not reach, however good the record is on both sides of it.

END OF THE DAY 21W

A reading is a value; a rate is a limit, and the limit is the part that says what happens next.

DAY 2 OF 15

The same water, moving faster

PLAN CARD 107W

Tuesday. The rain has started on the upper catchment. The gauge is climbing. The level is climbing behind it. Reggie Wilkes, the gate mechanic, has a question. Wind the gate open at a steady rate, and what happens to the discharge going out? That is a question about the spillway relation, not about the gate. Mercy Anand, the powerhouse chief, asks the same thing about power. Power leans on two quantities, and both of them are moving at once. Today you differentiate the three relations the dam is run from. A rate on the left picks up every part of the right side that is moving too.

BRIEFING 18W

Three formulas on three walls, and today every one of them is asked how fast it is moving.

WHAT YOU DECIDE 16W

Differentiate the three relations this dam runs on, and see what each rate carries with it.

2.1 Adding up a curve nobody wrote down

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · BALLPARK

SCENE 38W

Ekundayo has 5 readings an hour and a half apart — 48, 60, 74, 87 and 96 cubic metres a second. Sato is asking for the volume already in, and there is no equation behind any of it.

GUIDE 69W

The five numbers are all sums of the same readings, made in different ways. One halves the two ends before adding; one does not. One adds every reading flat. The rule the volume comes from decides which sum is the right one, so read the rule before the numbers. Counting an end reading at full weight adds water that never arrived, and the error grows with every extra interval.

ASK 9W

Estimate the volume that arrived over those six hours.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the area under a flow-against-time curve is a volume · a trapezium's area is the average of its two parallel sides times its width

VERDICT 91W

The fundamental theorem says the accumulated volume is the area under the flow curve. Nothing in it requires a formula, because a list of readings has an area too. The trapezoid rule averages each adjacent pair of readings. That is why every interior reading is used by two trapezia and each end reading by one. It is exact on any stretch where the flow runs in a straight line. On a rise that curves upward it reads slightly high, which is the direction to know before it goes to a committee.

TAKEAWAY 25W

The trapezoid rule adds an unwritten curve by averaging each pair of readings, so the interior readings are counted twice as heavily as the ends.

2.2 A gate is not a tap

SPILLWAY & GATES · GATE HOUSE · A PERSON · DERIVE

SCENE 38W

Wilkes has the spillway rating on the wall, $Q = C \cdot L \cdot H^{(3/2)}$ with $C \cdot L$ equal to 21.0. The head over the sill is 2.0 metres and rising, and the control room wants the discharge rate rather than the discharge.

GUIDE 50W

Build it a line at a time. The control room wants a rate of change of discharge, not a discharge. So this is a related-rates problem. Differentiate the rating with respect to time, and the chain rule brings dH/dt along. Evaluate what you derive at a head of two metres.

ASK 16W

Derive how fast the discharge rises as the head rises, and evaluate it at 2.0 metres.

BEHIND THE BUTTON 159W

Why the chain rule is what appears. Q depends on H, and H depends on time, so Q's dependence on time runs through H. Differentiating the rating therefore produces a factor for how Q responds to H. It also produces a factor for how fast H is moving. The second is the number the gauge is giving you.

Why a gate is not a tap. A tap's flow is proportional to how far it is open. A spillway's discharge goes as head to the power of three halves, so the same centimetre of level does more the deeper the water over the sill. The rating is nonlinear, and the operator's intuition has to be too.

What the control room does with a rate. A discharge tells you what is going out now. Its rate of change tells you how much time you have before it exceeds the channel downstream, and that is the number a warning is issued on.

TAKES AS READ

the discharge over a weir depends on the head raised to a power · a quantity that changes because another one changes needs the chain rule

VERDICT 93W

The exponent 3/2 is the whole behaviour of a spillway. The power rule brings it down as a factor and drops it to 1/2. So the sensitivity of discharge to head grows as the square root of the head. The last half metre does far more than the first. The chain rule then supplies dH/dt, because nothing here is a function of the head directly; it is a function of time through the head. Drop that factor and the answer is a slope with respect to the wrong variable, in units nobody checks.

TAKEAWAY 19W

The power rule brings the exponent down and reduces it by one; the chain rule supplies the inner rate.

2.3 Which quantity is which

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · PROTOCOL

SCENE 36W

Baptiste has four expressions on the briefing sheet, all built from the same inflow record. A committee this evening will read whichever number is nearest the top. She wants each one labelled before it goes out.

GUIDE 63W

Four expressions, all built from one inflow record, and the units separate them before any argument does. Integrating a flow multiplies a rate by time; differentiating divides by it. Read the units off each expression first, and the pairing follows. A committee reads the top figure on a sheet and cannot tell a volume from a rate. The label is the whole safeguard.

ASK 9W

Match each expression to the quantity it actually reports.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

integrating a rate over an interval gives a total, and differentiating a total gives a rate · the units of an expression follow from the units of the things it is built from · riemann sums, left and right endpoints, and the trapezoid rule — taken as read

VERDICT 107W

These four are built from one record and answer four questions, and the units separate them before any argument does. Integrating a flow multiplies cubic metres a second by seconds and returns cubic metres. Dividing that back by the interval returns a flow again. That is the average value of the function. It is not the average of the readings, unless they happen to be evenly spaced. Differentiating once gives the

> climb and twice gives the curvature. Curvature is the quantity that says whether a flood is still accelerating. That is what the evening committee actually needs, and it is the one nobody puts on a sheet.
>
> **TAKEAWAY** 25W
>
> A rate, its accumulation and the rate of the rate are three different measurements of the same record, and only their units tell them apart.

END OF THE DAY 15W

Differentiating a relation converts it from a statement about values into a statement about rates.

DAY 3 OF 15

What fell, and what arrived

PLAN CARD 113W

Wednesday. The front arrived overnight. Ekundayo has six hours of gauge readings, one every fifteen minutes. Sato wants the volume of water that has already reached the reservoir. That is a total built out of a rate. There is no formula to work from, only a column of numbers. Down in the gallery, Halina Zawadzka, the structural engineer, has the same job in miniature. Her seepage weirs are reading higher each hour. She has to pick which end of each interval to trust, and that choice moves the answer. Today you add up a rate that exists only as a column of numbers. Then you say which way each method of adding leans.

BRIEFING 12W

Six hours of gauge readings and no formula behind any of them.

WHAT YOU DECIDE 20W

Add up a rate that only exists as readings, and say which way each way of adding it is wrong.

3.1 What the rain has to pass through

SPILLWAY & GATES · GATE HOUSE · A ROOM · CHAIN

SCENE 41W

Wilkes wants to know why the gate settings from this morning are already wrong. Berg, the flood engineer, puts the whole path on the board — rain, runoff, inflow, storage, level, head, discharge — and asks which step decides the answer.

GUIDE 71W

Put the transfers in the order the rate actually travels. That order is rain to runoff to inflow to storage to level to head to discharge. Then name the one step that decides how much of it reaches the sill. Every relation on the path contributes a factor, and a path with a step missing is not a path. The largest number on the board is not automatically the governing link.

ASK 22W

Put the transfers in the order the rate travels, then name the one that decides how much of it reaches the sill.

BEHIND THE BUTTON 169W

Why the whole path is a product. Each step is a relation with its own gain. How much rain becomes runoff. How much runoff arrives as inflow. How much inflow becomes stored volume. How volume becomes level. How level becomes head. Multiply the gains and you have how a millimetre of rain becomes a cubic metre a second at the sill.

What 'governing' means. Not the biggest term — the one whose value the answer is most limited by. A step that passes on nearly everything is not deciding anything, however large the numbers flowing through it. A step that throttles the rate sets what everything downstream can be.

Why the morning's settings went wrong. A gate schedule is written from an assumed path. One step's gain may have been assumed and be actually different. The catchment may be wetter than modelled. The level-to-head relation may be off by the sill's own geometry. Then every setting downstream of it inherits the error. That is what the board is for.

TAKES AS READ

a rate reaches a distant quantity only through every relation that lies between them · a chain of rates is limited by the link whose own derivative is smallest

VERDICT 103W

Every link on this path multiplies the rate by its own derivative, and the product is what arrives. The spillway link is the one everybody watches because its exponent makes it dramatic. But three- halves of a square root is a factor near 40. The storage-to-level link divides by a surface area of 2 and a half million. That single division is why a catchment can double its flow in a morning and the level move eight centimetres. It is also the link the resurvey is about to change, which is the reason it is worth naming now rather than in a week.

TAKEAWAY 24W

A related-rates problem is a chain, and the link with the smallest derivative governs how much of the rate arrives at the far end.

3.2 Which end of the interval you believed

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · CHOICE

SCENE 38W

Zawadzka has hourly weir readings that climbed all shift, and two totals on the board from the same column of numbers. One used each hour's opening reading, the other its closing one, and they differ by a tenth.

GUIDE 81W

The four answers differ in how confident they are willing to be. Two rank the totals against the truth, one refuses to rank them, one calls a total correct. What settles it is a fact already on the board: the seepage rose all shift. Ask what a rectangle built on a rising hour's first reading must do, and what one built on its last must do. Two totals that bracket the truth are worth more than one that is nearly right.

ASK 12W

What can be said about the two totals without any further readings?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

a rate multiplied by an interval gives the accumulation over that interval only if the rate is constant · a sum of rectangles approximates the area under a curve

VERDICT 91W

Each rectangle is one hour wide and as tall as the reading it was built from. On an hour where the seepage only rises, the opening reading is the lowest value the rate took. That rectangle sits entirely under the curve. The closing reading is the highest, so its rectangle sits entirely over it. The rise is the whole argument, and it brackets the answer with no further measurement. That is worth more than either total, because it turns two disagreeing numbers into an interval the truth has to lie in.

TAKEAWAY 24W

> For a rate that only rises, a left-endpoint sum is always short and a right-endpoint sum always long, and the truth is between them.

3.3 The release she would have made

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A PERSON · DERIVE

SCENE 39W

Berg has this afternoon's schedule as $Q(t) = 48 + 8t$ cubic metres a second, running 6 hours. She wants the flat release with the same total, to set beside the one she asked for on the first day.

GUIDE 67W

Build it a line at a time. The steady flow with the same total is the ramp's average over the six hours. One way to get it survives a schedule that is not a straight line. Integrate the ramp and divide by the interval. Finish with a flow in cubic metres a second, so it can be set beside the one she asked for on day one.

ASK 13W

Derive the steady flow that delivers the same six-hour total as the ramp.

BEHIND THE BUTTON 180W

What a mean value of a function is. The average of Q over an interval is the integral of Q across it, divided by the length of the interval. That is the height of the rectangle with the same area as the curve. For a straight ramp that is the midpoint value, which is why this one can be checked in your head after the working is done.

Why the equivalence is worth writing down. A ramp and a flat release with the same total deliver the same water. They do very different things to the channel below. One arrives gently and ends high. The other is constant. Being able to state the flat equivalent is how two schedules get compared on anything other than their shape.

What day one was about. The release Berg asked for on the first day was chosen without this arithmetic. Setting the equivalent flat flow beside it settles one question. Does the schedule that has been argued about all fortnight actually differ in the water it moves? Or only in how it moves it?

TAKES AS READ

a definite integral is evaluated as the difference of an antiderivative at the two ends · an average value is an accumulation divided by the length of its interval

VERDICT 91W

Two steps and one trap. Integrating gives the total in the mixed units the formula was written in. Dividing by the interval turns it back into a rate. That is the definition of an average value. It works for any schedule at all. What does not work for any schedule is averaging the two ends. That happens to agree here, because a straight line spends as long above its midline as below. Carry that habit onto tomorrow's curved release and the published figure is wrong in the direction the curvature points.

TAKEAWAY 26W

Reducing a schedule to one steady rate is integrating it and dividing, and the two ends only average correctly because the schedule is a straight line.

3.4 The slope between two readings — Review

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · CALLBACK · BALLPARK

SCENE 43W

June Sato, the forecast liaison, has the Braith tributary gauge beside the Ashfell one — 62 cubic metres a second at noon, 47 at 3. She is drafting the recession note and wants the rate of fall stated, not the two readings quoted.

GUIDE 69W

Five tiles and only three of them are ends of an interval. The order the 2 readings are subtracted in decides the sign, and the sign is most of what this number reports. A sum of the readings has thrown the difference away, and the clock hour is a label rather than a length. Subtract the wrong way round and the note tells a committee the river is climbing.

ASK 14W

Estimate the average rate of change of the Braith gauge across those 3 hours.

BEHIND THE BUTTON 108W

Where the direction lives. The magnitude of a slope survives being subtracted backwards; the sign does not. Later minus earlier, over later minus earlier, is the only arrangement that puts the river's direction into the answer. It is the arrangement worth doing by habit rather than by thought.

What a falling limb does to a straight-line carry. A recession bends: the fall is quickest just after the crest and slackens for days afterwards. An average slope taken across the steep part and projected forward therefore empties the reservoir faster on paper than the river will empty it. That is the opposite of the error the rising limb produces.

TAKES AS READ

an average rate of change is the change in a quantity divided by the change in time · subtracting the earlier reading from the later one is what makes a rate signed

VERDICT 83W

The slope of the line through 2 points does not care which way the river is going. Taking the later reading first puts the direction into the sign. A negative average rate is a fall of about 5 cubic metres a second every hour. Reverse the subtraction and the arithmetic still works and the units still read correctly, so nothing on the sheet objects. A recession also bends, so the fall slackens and this slope carried forward overstates how quickly the tributary empties.

TAKEAWAY 21W

A slope carries a direction, and the order the two readings are subtracted in is the only place that direction lives.

END OF THE DAY 22W

A rate known only at instants still accumulates, and every rule for adding it is wrong in a direction you can name.

DAY 4 OF 15

Two days of not deciding

BEFORE THE DAY — FOLLOW

The gate gallery round, walked once

The gallery is dark, wet and loud, and the round has a fixed order because half of it cannot be done while a gate is moving. Stay with whoever is walking it. The stops are where the order matters, and somebody who has only read the list will do them in the order they are written rather than the order they are possible.

PLAN CARD 110W

Thursday. On Monday, Sunniva Berg asked for a release and did not get one. The water that would have gone out at 120 cubic metres a second now has to go out at 480. It goes into a river with people living beside it. This time the schedule is a formula, not a table. So the volume it delivers can be worked out exactly, instead of summed piece by piece. Today you use the fundamental theorem of calculus in both directions. One way, an antiderivative turns a rate into a total. The other way, a total hands a rate back. Berg wants to know what two days of waiting cost.

BRIEFING 20W

The release that was deferred on Monday has to be made today, and the arithmetic for it is an integral.

WHAT YOU DECIDE 16W

Use an antiderivative to evaluate an accumulation, and say what the accumulation function's own rate is.

4.1 What the level rise leaves out

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · BALANCE

SCENE 35W

The reservoir gained 1.06 million cubic metres overnight. Berg has the inflow record beside it. She wants the total that arrived at the dam, not the part that stayed, before she sizes this afternoon's release.

GUIDE 68W

Reading a stream costs nothing, so read them all. Then count only the ones that belong in the total she asked for. That total is the volume that arrived at the dam over twelve hours, not the part that stayed. The level rise is what remained after everything that left, so anything already leaving has to go back into the sum. Report the total when the ledger closes.

ASK 15W

Read what you need and report the total volume that arrived over the 12 hours.

BEHIND THE BUTTON 170W

Why the rise is not the arrival. The reservoir gained 1.06 million cubic metres, and it gained that while the turbines, the spillway and the compensation flow were all taking water out. What arrived is what stayed plus everything that left, which is why a balance is a ledger rather than a reading.

What counting a stream claims. Reading is free and neutral; adding a stream to the total says that this water is part of the quantity being asked for. Some streams on the board are not. A gauge upstream that also appears in the inflow record would be counted twice. A rainfall figure over the reservoir surface may already be inside the level rise.

Why Berg wants arrivals before sizing a release. The release has to be set against what the catchment is actually delivering. Not against what the reservoir happened to keep while the machines were running. Using the rise would size this afternoon's release against a number that already had this morning's releases baked into it.

TAKES AS READ

what accumulates is what is left after everything already leaving has left · a rate that runs throughout an interval contributes its rate times the whole interval

VERDICT 96W

The storage change is the integral of inflow minus outflow, so recovering the inflow means integrating every outflow term back in. Two of them are easy to miss for the same reason — nothing on the panel moves when they run. The machines and the seepage are steady rates that were on for the whole 12 hours. A steady rate integrates to rate times time without any further work. Report the 1.06 and the catchment's yield is understated by half. That yield is the number every forecast for the rest of the week is scaled from.

TAKEAWAY 23W

> A net change is the integral of a difference, so every removal term has to be added back before the arrival is known.

4.2 The schedule as a formula

SPILLWAY & GATES · GATE HOUSE · A PERSON · DERIVE

SCENE 39W

Wilkes has this afternoon's ramp written as $R(t) = 120 + 60t$ cubic metres a second, with t in hours from 2 o'clock. It runs for four hours and Berg wants the volume it will deliver before it starts.

GUIDE 55W

Build it a line at a time. A volume is the integral of a flow over time. So the ramp's total is an integral of a linear function. The units have to be reconciled on the way. The flow is per second and the schedule is in hours. Finish with the volume in cubic metres.

ASK 10W

Evaluate the volume the four-hour ramp delivers, in cubic metres.

BEHIND THE BUTTON 152W

Why integrate a straight line at all. The average of a linear ramp is its midpoint value, so this total can be had geometrically as the area of a trapezium. Doing it as an integral is worth the extra line. The same working survives a schedule that is not a straight line. Most real ones are not.

Where the units bite. R is in cubic metres a second and t is in hours, so integrating without converting gives an answer 3,600 times too small. The safe habit is to put the conversion into the integrand rather than remembering it at the end, where it looks like a factor somebody invented.

Why Berg wants it before it runs. A four-hour release is a commitment of water that cannot be recalled. Knowing the volume beforehand is what lets it be checked against what the reservoir can spare and what the channel below can take.

TAKES AS READ

the accumulation of a rate over an interval is the definite integral of that rate · an antiderivative of a power of t raises the exponent by one and divides by the new exponent

VERDICT 87W

The fundamental theorem is what makes this a subtraction rather than a sum. Any antiderivative works, because the constant appears at both ends and cancels. That is why a definite integral is a number while an indefinite one is a family. What costs people the answer is not the calculus but the bookkeeping under it. The rate is measured per second and the variable is counted in hours. So the integral comes out in mixed units, and it has to be converted once, at the end, deliberately.

TAKEAWAY 25W

The fundamental theorem evaluates an accumulation as the difference of an antiderivative at the two ends, so no sum is needed when a formula exists.

4.3 The curve that is already the answer

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · CHOICE

SCENE 40W

Ferrand keeps a running total of everything that has arrived since Monday, plotted as one rising curve. Rasmussen, the survey engineer, asks what its slope at any point is, and whether that is a new measurement or an old one.

GUIDE 69W

Every option here is a real quantity of this record. They differ in whether they name a total, a rate, an average, or a rate of a rate. A slope is a rate, so whatever answers this has to carry the units a rate carries. If the curve and the gauge are two views of one record, a disagreement between them is a bookkeeping error and not a flood.

ASK 13W

What is the slope of the running-total curve at four o'clock on Wednesday?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

an accumulation function reports the total of a rate from a fixed start up to a moving end · a derivative reports the slope of a curve at a point · the definite integral as an accumulation, and its units — taken as read

VERDICT 99W

The running total is an accumulation function, with a lower end fixed on Monday and an upper end that moves. Push that upper end forward by a small interval and it gains the volume arriving during it. That volume is the inflow times the interval, so the slope is the inflow itself. This is the first part of the fundamental theorem, and its use here is practical. It means the gauge and the total curve are two views of one record. A total curve that steepens where the gauge did not is a bookkeeping error rather than a flood.

TAKEAWAY 25W

The slope of an accumulation curve at any instant is the rate that was arriving at that instant. Differentiating an integral returns what was integrated.

4.4 Adding up a curve nobody wrote down — Review

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · CALLBACK · CHOICE

SCENE 40W

Ayo Ekundayo, the catchment hydrologist, has 5 readings off the overnight tape, and the recorder skipped twice. Two of the gaps are 90 minutes and 2 are half an hour. Sato has already totalled the column with the half-ends rule.

GUIDE 61W

The 4 answers disagree about what the half-ends form actually is. Is it a weighting of readings? Or a shortcut that quietly assumed something about the clock? Ask what a single trapezium is made of before arguing about weights. Getting this wrong does not shift the total slightly. It multiplies most of the night by a width the tape never had.

ASK 14W

What is wrong with Sato's total, and what does it take to put right?

BEHIND THE BUTTON 118W

What the half-ends form is a shortcut for. Add the 4 trapezia of a 5-reading tape and collect the terms. Every interior reading turns up in two of them and every end reading in one. That is where the halves come from. That collection is only legal when the widths are equal, because it is the common width that was factored out.

Why the fix is cheap. Nothing about a trapezium needs its neighbours to match it. Average the pair, multiply by the gap that separates that pair, and add the 4 results. It is more arithmetic than the shortcut and exactly as exact, which is why an uneven tape is an inconvenience rather than a lost measurement.

TAKES AS READ

the area under a flow-against-time curve is a volume · a trapezium's area is the average of its 2 parallel sides times its width

VERDICT 84W

The trapezoid rule is a sum of trapezia, and the half-ends form is what that sum collapses to when every width is the same. Take the widths apart and the shortcut is gone. Each adjacent pair is averaged and multiplied by the gap that separates it, and 5 readings give 4 areas to add. The uneven tape costs the arithmetic and nothing else. A 90-minute gap totalled as if it were half an hour reports a third of the water that arrived across it.

TAKEAWAY 21W

The half-ends form is an arithmetic shortcut that assumed equal spacing, and the trapezia are what survive when the spacing goes.

END OF THE DAY 19W

An antiderivative turns an accumulation into a subtraction, and an accumulation differentiates back to the rate that built it.

The last half metre

PLAN CARD 115W

Friday. The head of water over the spillway sill has climbed into a new part of the rating curve. Up here the last half metre does more work than the first two did. Bo Ferrand cannot rearrange the stage to storage curve into a neat formula for the level. That does not stop the level from having a rate. In the gallery, the uplift gauges are tied to the reservoir by a relation with the unknown on both sides. Anand wants the same arithmetic done for the machines. Today you differentiate a relation without solving it first. Then you follow one rate the whole way, from rain on the hills to water past the sill.

BRIEFING 18W

Two quantities tied by a curve nobody can rearrange, and one rate that has to travel four steps.

WHAT YOU DECIDE 15W

Differentiate a relation that will not solve, and follow a rate through everything it passes.

5.1 Why the same inflow moves the level differently

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · BALLPARK

SCENE 35W

Ferrand has the net inflow at 28 cubic metres a second and the surface area at this level at 2.6 square kilometres. Halloran wants tomorrow morning's level, which means a rate and not a volume.

GUIDE 69W

The numbers offered describe two different states of the reservoir and two different quantities. One area belongs to this level; another belongs to winter drawdown. One flow is the net figure; another is what one machine passes. A relationship is only true of the state it was written for. Using an area from a lower level makes the level rise faster on paper than it does in the valley.

ASK 15W

Estimate how fast the level is rising with 28 cubic metres a second net inflow.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

a volume rate divided by a plan area is a rate of level · the surface area of a reservoir depends on the level it is holding

VERDICT 90W

The relation is $dh/dt = (I - O) \div A(h)$, and the part that makes it calculus rather than division is that A depends on h. The same net inflow buys less height the higher the reservoir gets. So the level rate falls while nothing about the water changes. A straight-line forecast made this morning drifts high by the afternoon. It is also the first appearance of the term that the resurvey is going to move. Every level forecast on this site divides by an area somebody measured in 2003.

TAKEAWAY 25W

A level rate is a volume rate divided by the area it is spread over, and that area is itself a function of the level.

5.2 The curve that will not rearrange

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A PERSON · DERIVE

SCENE 36W

Zawadzka has the foundation relation as $U^2 + 0.8 \cdot U \cdot h = 640$, tying uplift pressure to reservoir level. Nobody has ever solved it for U, and she wants the rate of uplift while the level is climbing.

GUIDE 48W

Build it a line at a time. The relation ties U and h together and cannot be rearranged for U. So differentiate it as it stands, implicitly, with respect to time. Both dU/dt and dh/dt appear. Solve the result for dU/dt and evaluate it at the stated values.

ASK 18W

Derive dU/dt from the relation as it stands, and evaluate it at $U = 20$, $h = 25$.

BEHIND THE BUTTON 173W

Why implicit differentiation is not a trick. Every term is a function of time because U and h both are. Differentiating both sides therefore requires the product rule on the $U \cdot h$ term. It requires the chain rule wherever U appears. The outcome is one linear equation in dU/dt. That can be solved without ever solving for U.

Why this relation resists rearranging. $U^2 + 0.8Uh = 640$ is quadratic in U with a coefficient that depends on h. So U can be written in closed form only through the quadratic formula and a square root. That square root must be tracked for sign. Differentiating first avoids all of that and gives the rate directly.

Why uplift is what she is asking about. Water working under a dam pushes upwards, and that pressure is subtracted from the weight holding the structure down. A rate of uplift as the reservoir climbs is therefore a rate at which the safety margin is being spent. That is why it is wanted while the level is still rising.

TAKES AS READ

a relation between two quantities can be differentiated without being solved · a product of two quantities that both change contributes two terms

VERDICT 85W

Implicit differentiation exists for exactly this case. The relation ties two quantities without expressing either as a formula for the other. Differentiating both sides as they stand costs nothing. Every appearance of the dependent quantity carries its own rate by the chain rule, and that is the step that gets dropped. What comes out is a sign nobody would guess from the story. The uplift is falling while the reservoir climbs, because the relation is not a straight line, and only differentiating it says so.

TAKEAWAY 20W

Implicit differentiation gets a rate out of a relation that cannot be rearranged, by differentiating both sides as they stand.

5.3 What the machines make of a rising head

POWER & DEMAND · POWERHOUSE · A ROOM · BALLPARK

SCENE 40W

Anand has the head on the machines rising at 0.31 metres a day with both units passing a steady 46 cubic metres a second. She wants the rate of change of output for the grid schedule, in kilowatts a day.

GUIDE 72W

Five numbers, and the trap is that all of them are true of the machines this morning. Two are rates and three are standing values. The product rule needs each rate paired with the right factor, so ask of every number whether it is a level or a speed. A rate that is zero today is still a term in the relationship. It stops being zero the moment somebody touches a gate.

ASK 12W

Estimate how fast the machine output is rising, in kilowatts a day.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

a product with one factor held constant differentiates to the constant times the other rate · a rate of change carries the units of the quantity divided by the units of time

VERDICT 91W

The output is 8.6·Q·H and the flow is not moving, so the first term of the product rule is 0 and the entire rate is 8.6·Q·(dH/dt). What survives is a multiplication. There is still a reason to write the product rule down. Today's 0 is an accident of the gate settings. The moment anybody touches a gate the other term reappears. Then this number is wrong by more than it is right. It is also the rate that pays for the fortnight, which is why Halloran asks for it every morning.

TAKEAWAY 23W

With one factor of a product held fixed, the whole rate comes from the other, scaled by the constant that was standing still.

5.4 The level, through a wet afternoon

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · HOLD

SCENE 34W

The reservoir has to sit at its operating level while the catchment delivers whatever the afternoon brings. The gates are the only outflow anybody controls, and the inflow is decided a long way upstream.

GUIDE 54W

Hold the level inside the band on the panel. The band narrows as the afternoon goes on, because the freeboard above the spillway is what the whole scheme is protected by and it does not grow. The gate opening is your control, and every change in inflow keeps arriving until the outflow matches it.

ASK 10W

Hold the reservoir at its operating level through the afternoon.

BEHIND THE BUTTON 142W

Why the level is an accumulation. The rate of change of storage is inflow minus outflow, so the level is the integral of that difference. A gate opening that does not match the inflow does not move the level to a new place. It changes how fast it is moving. That is the whole difficulty.

Why the inflow will not sit still. Rain upstream arrives as a wave hours later, a tributary joins after the second band of weather, and the catchment goes on delivering after the rain has stopped. None of it is a step, and none of it is negotiable.

Why the margin shrinks. Freeboard is fixed. The longer the afternoon has run, the more of it has already been spent. So the same rate of rise is a larger share of what is left at four than at one.

TAKES AS READ

water stored is what has come in minus what has gone out

VERDICT 155W

The rate of change of storage is inflow minus outflow. So the level is the accumulation of that difference. That is why holding it feels unlike holding anything with a setpoint. A gate opening does not put the level somewhere; it decides how fast the level is moving. Match the inflow and the level stops wherever it happens to be. Fall short and it rises at the difference for as long as the mismatch lasts. That is the same integral the whole course keeps returning to, drawn in real time on a gauge. The disturbances are the catchment doing what catchments do. A rain wave arrives hours after the rain. A tributary joins late. The ground goes on delivering once the sky has cleared. And the band narrows because freeboard is fixed and finite. The margin above the spillway is what the scheme is protected by, and none of it comes back during the afternoon.

TAKEAWAY 21W

> A level is the accumulation of a difference, so it is held by matching rates rather than by chasing a number.

END OF THE DAY 20W

A relation can be differentiated where it cannot be rearranged, and a rate travels through every relation between its ends.

DAY 6 OF 15

Has it peaked?

PLAN CARD 112W

Saturday. The first good news of the fortnight is on Ekundayo's screen. The inflow is still rising. The hourly steps up are not getting bigger any more. Nobody will take that as a hunch. A rate that has peaked and a rate that is merely easing off make for two very different afternoons. Ferrand has the level fitted to a curve that turns over on a day you can name. Baptiste has to warn the reach, and she has to say which peak she means. The two peaks do not fall on the same day. Today you find the point where a climb turns over. Then you say which quantity peaks when.

BRIEFING 21W

The gauge is still climbing, and the question on the call is whether it is climbing as hard as it was.

WHAT YOU DECIDE 13W

Find the inflection in a rising record, and say which quantity peaks when.

6.1 The hour the increments turned

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · PROBE

SCENE 57W

Ashfell's gauge reports the flow once an hour, and it has risen at every reading so far. Rising is not the question. Sato wants the hour when the climb turned over — the flow still going up, but going up by less than it did the hour before. She will not take a shape described in words.

GUIDE 57W

Nothing is read for you. Press Read on any hour and it reports that hour's flow, the climb since the hour before, and how that climb compares with the one before that. Take at least three, and consecutive hours tell you more than scattered ones. Then click the hour where the climb stopped getting steeper, and commit.

ASK 16W

Read as many hours as you need. At which reading did the climb stop getting steeper?

BEHIND THE BUTTON 186W

What the two columns are. The climb between consecutive readings is an estimate of the first derivative — how fast the flow is changing. The change in those climbs is an estimate of the second derivative, which is how fast the changing is changing. Both are estimates rather than values, because an hourly gauge cannot see what happened between two readings.

Why an inflection is worth an hour of anybody's morning. The flow is still rising at the turn, so nothing about the number itself announces it. What has changed is the trend. From that hour on, each hour adds less than the last. That is the first evidence that a flood peak exists and is approaching. Waiting for the flow to fall means waiting until the peak has already passed.

Why consecutive readings matter. A difference of differences amplifies noise: two readings each a little uncertain give a climb that is more uncertain, and comparing two climbs more uncertain again. Reading hours next to each other keeps the comparison honest, where jumping about the log invites reading a gauge wobble as a change of shape.

TAKES AS READ

the change between consecutive readings estimates the first derivative · the change in those changes estimates the second derivative

VERDICT 88W

Every reading in this column is higher than the one before it. So the first derivative never changes sign. The flow does not peak anywhere on the screen. What changes is the second derivative: the increments grow to 18 and then shrink. That turn is the earliest thing on a hydrograph that says the event has a top. It arrives hours before the top does. That is the whole reason to look for it. Reading only the flow leaves nothing to say until the peak has already passed.

TAKEAWAY 19W

An inflection is where the second derivative changes sign, and it happens while the quantity itself is still climbing.

6.2 Where the curve turns

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · DERIVE

SCENE 37W

Ferrand has the level fitted to $h(t) = 214.6 + 0.9t - 0.03t^2$, with t in days from Monday. Halloran wants the day it stops rising and will not take a shape off a screen for it.

GUIDE 69W

Build the working down a line at a time. Finding where the level stops rising is finding where its rate is zero. But that is only half the job. A zero rate is a turning point. This one has to be shown to be a maximum rather than a minimum. Halloran will not take a shape off a screen, so the classification has to come out of the working.

ASK 18W

Find the day the level stops rising, and establish that it is a maximum rather than a minimum.

BEHIND THE BUTTON 165W

Why a zero rate is not yet an answer. The derivative is zero at a peak, at a trough and at a shelf. Solving $h'(t) = 0$ tells you where something happens; it does not tell you what. On a reservoir level the difference between the two is the difference between standing down and evacuating.

Two ways to settle it, and what each costs. The second derivative at the critical point is quickest: negative means the slope is falling through zero, which is a maximum. Checking the sign of the first derivative either side is slower and works when the second derivative is zero. Both are legitimate; the working has to show one.

Why a picture is not enough. A plotted curve with a flat top is convincing and unquotable: the eye cannot separate a maximum from a long shelf, and the screen's scale can hide either. A classification that follows from the algebra is the same claim in a form somebody else can check.

TAKES AS READ

a quantity stops rising where its derivative is zero · the sign of the second derivative distinguishes a maximum from a minimum

VERDICT 92W

Setting the derivative to zero finds where the level stands still. On its own that does not say whether it is a crest or a trough. A reservoir topping out on day 15 and One bottoming out on day 15 have the same first derivative there. The second derivative settles it with no further data. Because it is a constant here, it settles it on every day at once. A curve concave down over the whole fortnight has at most one turning point, so the day is unique as well as classified.

TAKEAWAY 17W

A critical point says where a quantity turns; only the second derivative says which way it turns.

6.3 The two peaks are not the same hour

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A PERSON · CHOICE

SCENE 37W

Baptiste has the inflow curve and the level curve on the same axes, peaking at different hours, and a reach to warn. Etta Prowse, the riverside landowner representative, wants to know which one the warning is about.

GUIDE 66W

All four options name a moment that is genuinely visible on the traces. They differ in which curve is being watched: the inflow, the outflow, or the difference between them. Storage answers to the difference and to nothing else. Warning a reach on the strength of an upstream peak is wrong in the expensive direction. The highest water arrives after the news that it would not.

ASK 9W

When does the reservoir level reach its highest point?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

the level rises whenever inflow exceeds outflow and falls whenever it does not · a quantity turns over where its own rate of change passes through zero

VERDICT 92W

The level is an accumulation and the net flow is its derivative. So the level turns over exactly where inflow minus outflow crosses zero. At the inflow peak the net flow is still strongly positive. The river is delivering more than the dam is passing, and the reservoir goes on filling for hours, usually most of a day. Telling the reach the peak has passed on the strength of an upstream gauge is wrong in the direction that costs the most. The highest water arrives after the news that it would not.

TAKEAWAY 23W

The reservoir peaks when the net flow is zero, which is later than the inflow peak and has nothing to do with it.

6.4 Which end of the interval you believed — Review

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · CALLBACK · PROTOCOL

SCENE 39W

Dr. Halina Zawadzka, the structural engineer, has 4 shifts of weir readings on the gallery board. The same 2 totals stand against each: one built on opening readings, one on closing. No shift behaved like the one before it.

GUIDE 61W

Four shifts and 4 verdicts, one each. What settles every pairing is the direction the seepage moved rather than how large the readings were. Take the 2 shifts that only go one way first, and the pair left over separates on whether the record turns. A total nobody can rank is worth less than 2 that bracket the answer between them.

ASK 10W

Match each shift's record to what the two totals do.

BEHIND THE BUTTON 120W

Why the direction is the whole of it. A rectangle one hour wide and as tall as the opening reading sits under the curve on a rising hour and over it on a falling one. The closing-reading rectangle does the reverse. Nothing else about the readings enters, which is why a shift can be ranked before its numbers are read.

What a bracket is worth. Two totals that are known to sit either side of the truth are a statement with a width. The width is a measurement of how coarse the hourly tape is. One total that is nearly right and cannot be placed is not, which is the argument for keeping both rather than picking the better-looking one.

TAKES AS READ

a rate multiplied by an interval accumulates exactly only where the rate holds still · a sum of rectangles approximates the area under a curve

VERDICT 103W

A rectangle built on an hour's opening reading sits under the curve wherever the rate is rising. It sits over the curve wherever the rate is falling. The closing-reading rectangle does the opposite. So the direction of the record decides which total is short, and the size of the readings never enters. A shift that only rises is bracketed one way and a shift that only falls the other way. A shift with a peak in it is bracketed by neither, because each total is short across half the shift and long across the rest. A flat shift has no error at all.

TAKEAWAY 22W

Which endpoint total is short is decided by the direction of the record, and a record that turns is bracketed by neither.

END OF THE DAY 21W

The first derivative says which way, the second says whether the way is easing, and they turn over on different days.

DAY 7 OF 15

What the wall is carrying

PLAN CARD 106W

Sunday. Three people want a largest value. Zawadzka has an overturning moment on the wall. It grows as the uplift under the base spreads, and then it stops growing. Wilkes wants the place on the rating curve where the spillway is most sensitive. His head range is fixed by how far the gates open. Anand has a grid schedule to write. Her penstock loses head the harder it is driven. So pushing more water through the machines stops buying more power somewhere in the middle of the range. Today you find three largest values. The three answers do not sit in the same kind of place.

BRIEFING 14W

Three quantities that are largest somewhere, and the somewhere is different in each case.

WHAT YOU DECIDE 15W

Find a maximum at a critical point, at an endpoint, and by sweeping a control.

7.1 The worst extent of the uplift

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · BALLPARK

SCENE 34W

Zawadzka has the overturning moment about the heel as $M(a) = 12a - 1.5a^2$, in meganewton metres per metre of wall. Here a is the extent of the uplift under the base, in metres.

GUIDE 66W

The numbers offered are all coefficients from the moment expression, some before differentiating and some after. Work the derivative out on paper first, then match its coefficients against what is on the table. Two of these belong to the moment itself and never appear in its derivative. Taking a coefficient straight from the moment gives a clean-looking answer that puts the worst case outside the wall.

ASK 11W

Estimate the uplift extent at which the overturning moment is greatest.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

a quantity is largest where its derivative is zero, if it has such a point inside the range · the power rule brings an exponent down as a factor and reduces it by one

VERDICT 84W

Differentiating gives $12 - 3a$, and the 3 is the whole answer. The exponent 2 comes down as a factor and multiplies the 1.5 that was already there. Using the 1.5 instead puts the worst case at 8 metres. That is past the downstream toe and outside the wall entirely. It is an answer that is not merely wrong but physically impossible. Nobody catches it, because the arithmetic is clean. The moment there is 24 against a true worst of 24 at 4 metres.

TAKEAWAY 25W

A critical point is found by differentiating and solving, and the factor the power rule brings down is the part that decides where it lands.

7.2 A maximum with no critical point

SPILLWAY & GATES · GATE HOUSE · A PERSON · DERIVE

SCENE 32W

Wilkes has yesterday's sensitivity, $dQ/dH = 31.5 \cdot H^{(1/2)}$, and a head range the gates hold between 0.5 and 3.0 metres. He wants the head at which a centimetre of level does the most.

GUIDE 60W

Build it a line at a time. The sensitivity rises with head and never turns over. So there is no critical point to find. The greatest value has to be at an end of the range the gates hold. Say which end and say why, because 'the derivative is zero somewhere' is the answer this stop is written to catch.

ASK 15W

Establish where on the range the sensitivity is greatest, and say why it is there.

BEHIND THE BUTTON 163W

Why there is no critical point. dQ/dH goes as the square root of H, which is increasing everywhere on 0.5 to 3.0 metres. Its own derivative is never zero on that interval. So the usual routine — set the derivative to zero, solve, classify — produces nothing at all. The extreme value is still there; it is just on the boundary.

What the theorem actually promises. A continuous function on a closed interval attains a maximum and a minimum. It does not promise they are interior. Candidates are the critical points *and* the endpoints, and half the exam marks lost on this topic are lost by checking only the first list.

What it means at the sill. The deeper the water over the crest, the more a centimetre of level is worth in discharge. So the gate operator's attention should be highest when the head is highest. That is the opposite of the intuition that a nearly empty spillway is the twitchy one.

TAKES AS READ

a continuous quantity on a closed interval attains a largest value somewhere on it · a critical point is where the derivative is zero or does not exist

VERDICT 85W

Looking for a critical point and finding none is a result, not a dead end. The sensitivity here is a square root with a positive coefficient. It climbs across the whole interval and never turns. Its own derivative is positive everywhere and merely gets smaller, which is the step easiest to misread as flattening off. On a closed interval a continuous quantity always attains a maximum. So once no interior candidate exists, the two ends are the only places left, and one of them wins.

TAKEAWAY 22W

A function with no critical point inside an interval still has a maximum, and it is at an end of the interval.

7.3 More water, less power

POWER & DEMAND · POWERHOUSE · A ROOM · SWEEP

SCENE 56W

More water through a turbine is more power, up to a point. As the flow rises the penstock loses head, and eventually less head per cubic metre beats more cubic metres. Anand has the machines on manual, and wants the flow to run them at tonight found on the instrument rather than argued at the table.

GUIDE 63W

Drag the flow. The readout is the output in megawatts, and higher is better; only the flows you stop at get plotted. The curve rises, turns over and comes back down, so the setting you want is somewhere inside the range rather than at either end of it. Sweep the whole range before deciding, then mark the flow you would run and commit.

ASK 14W

Move the flow and watch the output. Where should the machines be run tonight?

BEHIND THE BUTTON 169W

Where the head goes. Friction in the penstock rises roughly as the square of the flow, so doubling the flow costs four times as much head. The turbine is paid in head times flow, which means the two terms fight: one grows steadily, the other is eaten faster and faster.

Why the answer is in the interior. When two effects pull opposite ways and one accelerates, the total has a maximum somewhere in the middle. At that maximum neither effect is at its own best. This is the shape behind most operating points in the plant. It is the reason a plant has an operating point at all rather than a throttle you open fully.

What the top of the curve is like. Near the maximum the output changes slowly with flow, which is why an operator can trade a little output for a steadier machine. Far from it the loss is quick: at the top of the range these machines make less than half of what they can.

TAKES AS READ

the head available to a turbine falls as the flow through it rises · a quantity that rises and then falls has a largest value somewhere between

VERDICT 88W

Output is flow times head, and the head is itself falling with the square of the flow, so the product climbs, flattens and turns over. Differentiating $8.6Q(61 - 0.006Q^2)$ gives $8.6(61 - 0.018Q^2)$, which is 0 near 58 — the same place the trace turns, found by calculus instead of by hunting. The shape is the lesson. Near the top the curve is almost flat. Being a few cubic metres a second out costs almost nothing there. The same error low on the curve costs a great deal.

TAKEAWAY 24W

Two effects pulling opposite ways put the best setting in the interior of the range, where neither of them is at its own best.

7.4 The instant the gate moved — Review

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · CALLBACK · CHOICE

SCENE 41W

Nadine Baptiste, the downstream warning officer, has Thursday's tailrace trace. Gate 2 was wound open across 4 minutes and then stopped dead. The discharge climbs the whole way and holds after. She wants the instant the winding stopped read off it.

GUIDE 66W

The 4 answers mix 2 separate questions together: whether the trace breaks, and whether it has a slope. Settle the break first, from the readings either side of that instant. Then ask the slope question again from the left and from the right, as a second question with its own answer. Calling a corner a break sends a fault report about a gauge that is working.

ASK 12W

What is true of the discharge at the instant the winding stopped?

BEHIND THE BUTTON 110W

Two limits, not one. The limit of the discharge across an instant asks what value the trace approaches; the limit of the difference quotient asks what slope it approaches. They are separately capable of existing, and a corner is exactly the case where the first one does and the second one does not.

Why a corner is common and a jump is rare. Water has mass, so a discharge cannot teleport, and every genuine jump in this record is a gauge or a logging artefact. A corner needs nothing so dramatic. A hoist reaching its set position is enough. The trace goes on being trustworthy on both sides of it.

TAKES AS READ

a limit describes what a quantity approaches from one side · a derivative is a limit of the difference quotient taken across the instant · limits, one-sided limits, and where a limit fails to exist — taken as read

VERDICT 87W

Nothing jumps here. Both one-sided limits of the discharge are the same number. So the trace is continuous at that instant, and a value can be read straight off it. The derivative is a different limit, of the difference quotient, taken across the instant. It has 2 answers. One is a steep climb approaching from the left. The other is a flat line approaching from the right. Two one-sided slopes that disagree leave the 2-sided one undefined. Continuity buys the value and it never buys the rate.

TAKEAWAY 23W

Continuity is a promise about the value and never about the slope, and a corner is where the difference between the two shows.

END OF THE DAY 27W

A maximum sits at a critical point, at an end of the interval, or nowhere — and which of the three is a question about the derivative.

The release that just clears it

BEFORE THE DAY — HUNT

Six piezometers, and four are reporting

Every reading that says the dam is behaving comes from an instrument, and two of them have gone quiet since the last flood. Until they are found and checked nobody knows whether the silence is a cable or a change in the ground, and those two mean very different things.

PLAN CARD 96W

Monday of the second week. The release schedule changes through the day. The liaison committee wants one figure for it. That sounds like rounding off. It is a definite integral. Berg needs the same figure for her own reason. A steady release with the same total is the release she would have made on the first day. That comparison is the whole of her argument with Halloran. Today you reduce a changing rate to the one steady rate with the same total. Then you say what that steady rate guarantees about the morning it came from.

BRIEFING 15W

A schedule that varies all day, and a committee that wants one number for it.

WHAT YOU DECIDE 18W

Reduce a varying rate to the single steady one with the same total, and say what that guarantees.

8.1 One number for six hours

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · BALLPARK

SCENE 34W

Ferrand has Wednesday's 6 hours totalled at 1,582,200 cubic metres and a committee sheet with 1 line on it for the morning. He wants the steady flow that would have delivered the same water.

GUIDE 69W

The numbers on offer include a volume, the interval written in two different units, and two sums of readings. Only one pair is a volume and the time it took. An average value is an accumulation divided by a length, so the divisor has to be in the units the answer is quoted in. Mixing hours into a figure quoted per second is an error the arithmetic never reveals.

ASK 16W

Estimate the steady flow that would have delivered the same volume in the same six hours.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the accumulation of a rate over an interval divided by that interval is a rate again · 6 hours is 21,600 seconds · the definite integral as an accumulation, and its units — taken as read

VERDICT 92W

An average value is an integral divided by an interval, and the division is what turns a volume back into a flow. It is not the mean of the readings unless they happen to be evenly spaced in time. On a gauge that reports on demand they often are not. That is the error the definition exists to prevent. What comes out is the flat line with the same area underneath it as the real hydrograph. That is exactly the object a committee sheet with one line on it is asking for.

TAKEAWAY 23W

The average value of a rate is the accumulation divided by the length of the interval, not the average of the readings taken.

8.2 Whether the river ever ran at it

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A PERSON · CHOICE

SCENE 39W

Prowse has the published figure for the morning and the gauge trace beside it, and points out one thing. No reading in the column is that number. Ekundayo has to say what the figure does and does not claim.

GUIDE 61W

All four answers concede that no written-down reading equals the average. They differ on what that leaves: some instant, one particular instant, two instants, or nothing at all. Continuity is the fact doing the work here, not the arithmetic of the mean. Whether a published figure describes the morning or is only an artefact of division is what this distinction decides.

ASK 12W

What follows from the morning having averaged 73 cubic metres a second?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

a gauge trace of a river is continuous, without jumps · a continuous quantity takes every value between any two it takes · limits, one-sided limits, and where a limit fails to exist — taken as read

VERDICT 90W

The average lies between the lowest and the highest values the flow took. A river's discharge is continuous, so it cannot get from 48 to 96 without passing through every number between. The average is therefore attained at at least one instant. That is what makes the published figure a statement about the morning rather than an artefact of arithmetic. It says nothing about when. On a rise that steepens the crossing is later than the middle of the interval, and on a record that turns over it happens twice.

TAKEAWAY 22W

A continuous rate attains its own average somewhere inside the interval, though not necessarily at any moment somebody happened to write down.

8.3 Two things moving at once

POWER & DEMAND · POWERHOUSE · A ROOM · DERIVE

SCENE 40W

Mercy Anand writes the machine output as $P = 8.6 \cdot Q \cdot H$, in kilowatts, with Q the flow through the turbines and H the head on them. Both are moving this afternoon, and the grid wants a rate of change of output.

GUIDE 52W

Build it a line at a time. Output depends on flow and on head, and both are moving this afternoon. So differentiating with respect to time needs the product rule. It produces a term for each. Evaluate with the afternoon's numbers, and notice that the two terms can pull in opposite directions.

ASK 14W

Derive the rate of change of output, and evaluate it with the afternoon's numbers.

BEHIND THE BUTTON 170W

Why the product rule is the whole story. P is a constant times Q times H. Its rate of change is therefore the constant times the sum of two products. The first is how fast Q is changing times the current H. The second is how fast H is changing times the current Q. Neither term is meaningful alone, and the grid is asking about their sum.

Why the two terms fight. Opening the gates raises Q and lowers H, because more flow through the penstock costs head. So dQ/dt is positive while dH/dt is negative. The output may rise or fall depending on which product is larger. That is the same trade the operating point on day three is chosen against.

Why the grid wants a rate. A generator's output is scheduled, and a deviation is priced. Knowing the rate at which output is drifting is what lets a schedule be met without chasing it. It is a number the operator can produce from two gauges and one derivative.

TAKES AS READ

a product of two quantities that both change has a rate from each of them · a constant factor passes through a derivative unchanged

VERDICT 97W

A product of two changing quantities has two rates in it. The product rule keeps them separate. One term has the flow moving and the head held. The other is the other way round. Today the first term is zero because nobody has touched a gate. That makes the whole rise a head effect. The machines are producing more because the reservoir is filling behind them. That is worth knowing before anybody credits the operators. Multiply the two derivatives together instead and the answer is not merely wrong, it is not power per unit time at all.

TAKEAWAY 19W

When two factors change at once the rate is the sum of two terms, each holding one factor still.

END OF THE DAY 21W

The average value of a rate is its accumulation divided by the interval, and a continuous rate attains it somewhere inside.

DAY 9 OF 15

The ledger that will not close

PLAN CARD 108W

Tuesday. The low-level outlet has been open since first light. The level is falling, and it is not falling at a steady rate. The head driving the outlet is the very thing that is going down. Ferrand wants the level as a function of time, not as a table. That means solving the equation instead of reading it off. Ekundayo has one gauge reading, six hours old. He is being asked to state a drain constant from it. That turns out to be two questions wearing one number. Today you solve the drawdown equation. Then you turn what it gives you into an hour somebody can act on.

BRIEFING 18W

The low-level outlet is open and the level is falling on a curve nobody has written down yet.

WHAT YOU DECIDE 12W

Solve the drawdown equation, and find what a single reading leaves undetermined.

9.1 The level that drains itself slower

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · DERIVE

SCENE 37W

Ferrand has the outlet behaving as $dh/dt = -k\cdot\sqrt{h}$, with h the head on it in metres and k about 0.10 an hour. He wants the level as a function of time, not another table of readings.

GUIDE 57W

Build it a line at a time. The relation gives the rate of change of the level in terms of the level itself. That is what a differential equation is. So the move is to separate the two variables and integrate both sides. Finish with the level as a function of time, not another table of readings.

ASK 12W

Solve the drawdown equation for the level as a function of time.

BEHIND THE BUTTON 168W

Why the outlet behaves this way. Flow out of an opening goes as the square root of the head above it, so a falling level drains more slowly the lower it gets. That is the physics in $dh/dt = -k\sqrt{h}$, and it is why the drawdown is not a straight line however steady the outlet is.

What separating variables means. Get every h on one side with the dh, and every t on the other with the dt. Each side is then an ordinary integral. The constant of integration is fixed by the level at the moment you started. That is why an initial condition is part of the answer rather than an afterthought.

Why a function beats a table. A table answers the times somebody happened to write down. A function answers when the level will reach the outlet. It answers how long a drawdown will take. It answers what happens if k is a little different. That is the difference between a record and a tool.

TAKES AS READ

a differential equation with the variables on separate sides can be integrated side by side · an antiderivative raises a power by one and divides by the new power

VERDICT 93W

Separating the variables is the whole method. The equation ties a rate to the quantity itself, and dividing through by the square root puts every h on one side and every t on the other. Each side can then be integrated on its own. What comes out is not the exponential everybody expects from a draining tank. The square root makes it a parabola, and a parabola reaches zero. That decides an operational question rather than a mathematical one, because this outlet finishes at a stateable hour and a decaying exponential never would.

TAKEAWAY 19W

Separating the variables turns a rate that depends on the quantity into two ordinary integrals, one on each side.

9.2 The reading that fits many curves

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A PERSON · DEGENERACY

SCENE 34W

Ekundayo has one level reading six hours into the drawdown and is being asked for the drain constant. Rasmussen has a second reading from 12 hours later and has not brought it over yet.

GUIDE 70W

Two controls move the predicted curve, and the readout is how well that curve matches the reading you have. One reading is not enough to pin both: you will find a whole family of settings that match it equally well. Slide along that family, see how wide it is, and only then bring in the second reading. Report the drain constant when you can say it rather than guess it.

ASK 12W

Match the six-hour reading, then say what the drain constant actually is.

BEHIND THE BUTTON 162W

Why one reading cannot do it. A drawdown curve has a size and a rate, and a single level six hours in constrains their combination rather than either one. Trade a larger initial volume against a slower drain, and the curve passes through the same point. That is what the family of settings you can slide along actually is.

What breaks the tie. A second reading twelve hours later is on a different part of the curve. There the same trade no longer holds. Settings that agreed at six hours disagree at eighteen. That is why the second observation is worth more than a repeat of the first, and why Rasmussen's undelivered reading is the whole stop.

Why this matters beyond the arithmetic. Reporting a drain constant from one reading is reporting a number the data does not contain. The honest answer at that point is the range, and the professional move is to go and get the reading that narrows it.

TAKES AS READ

a solved differential equation carries a constant that the equation itself does not fix · two unknowns need two independent measurements

VERDICT 98W

Integrating produced a constant, and the equation cannot supply it; only a measurement can. One reading is one equation in two unknowns. So a higher starting head paired with a faster drain reproduces it exactly. That is why the match stays perfect all along the curve. The second reading is not a repeat of the first. It is taken at a different time. So it constrains the same two unknowns in a different combination, and the two lines cross once. Sliding along a locus with a perfect match is what an underdetermined problem feels like from the inside.

TAKEAWAY 25W

A single reading fits a whole family of starting levels and drain constants, and only a second reading at a different time picks one out.

9.3 How long the drawdown takes

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · BALLPARK

SCENE 37W

Baptiste has the solved drawdown curve and a reach to warn. The head starts at 25 metres and the outlet has to be shut when it reaches 16, and she needs the hour rather than the shape.

GUIDE 77W

Five numbers are offered and two of them are heads rather than roots of heads. The solved equation is written in square roots, so a head dropped straight into it is the wrong shape. Test each number against the form of the equation before its size. The square roots are also where the surprise lives. The second nine metres of drawdown take longer than the first, so a warning carried forward in a straight line arrives early.

ASK 15W

Estimate how long the outlet takes to bring the head from 25 metres to 16.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the drawdown solution can be rearranged for the time at a given head · the square root of 25 is 5 and the square root of 16 is 4

VERDICT 90W

The solution rearranges to $t = 2(\sqrt{h_0} - \sqrt{h}) \div k$. That is the whole practical value of having solved it. A curve becomes an hour somebody can act on. The square roots are also where the surprise is. The first 9 metres of drawdown take 20 hours. The next 9 take 26. The head driving the outlet is falling the whole time. A warning built on the first nine metres and carried forward linearly arrives early, and being early is not free on a river with people beside it.

TAKEAWAY 18W

Rearranging the solved equation for time turns a curve into a number somebody can put on a warning.

9.4 Rising, or reading high

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · BELT

SCENE 33W

The morning report needs the night's readings sorted first. Some of what moved overnight is the dam and the water. The rest is the instrument, and both arrive as a number that changed.

GUIDE 46W

Two bins. Something real moves in a way the physics allows and other instruments agree with — a piezometer follows the level, a plumbline follows the temperature. An instrument fault moves in ways nothing corroborates: instantly, off scale, or not at all when everything else has.

ASK 11W

Send each overnight change to the bin that says what moved.

BEHIND THE BUTTON 77W

Why corroboration rather than plausibility. Dam instruments are redundant on purpose: two piezometers in one line, plumblines and geodetic survey measuring the same movement. A reading with no partner moving is a reading about the instrument.

What a real movement looks like overnight. Small, smooth, and correlated with something — the level, the temperature, the season. Concrete moves with temperature every day of its life, and a plumbline that did not move overnight is the surprising one.

TAKES AS READ

an instrument can fail while the thing it measures does not

VERDICT 148W

A dam is instrumented redundantly because a single number cannot be checked. Real movement is smooth, small and correlated. The piezometers follow the reservoir level with a lag. The plumblines swing with the daily temperature cycle. The seepage weir rises when the level does. A fault is none of those things. It steps instantly, parks past the end of its range, freezes while everything around it moves, or contradicts the second instrument in the same borehole. That is why the sort is by corroboration rather than by how alarming the number is. It is also why the frozen reading matters as much as the jumping one. Concrete moves with temperature every night of its life. So a plumbline that did not move is not reassuring. It is an instrument that has stopped reporting. And this is the one structure where nobody can afford to find that out later.

TAKEAWAY 14W

What decides whether a reading is real is whether anything else agrees with it.

END OF THE DAY 24W

A rate that depends on the quantity itself is solved by separating and integrating, and the constant that comes out needs its own measurement.

DAY 10 OF 15

Every number again

PLAN CARD 118W

Wednesday. Ilya Rasmussen has been out over the reservoir with an echo sounder. The stage to storage curve was surveyed in 2003 and never done again. Two decades of silt have settled since. So the surface area at any given level is smaller than the old sheet says. Nothing was faked and nobody was careless. But volume is the integral of area with respect to level. The area is the integrand, the thing being added up. Get it wrong by a few per cent, and every total this control room has worked out for a fortnight is wrong the same way. Today you rebuild a volume from the corrected areas. Then you find how far that correction travels.

BRIEFING 18W

The survey boat has been out, and the curve every volume on this site came through is wrong.

WHAT YOU DECIDE 14W

Rebuild the volume from the area curve, and find which quantity the correction reaches.

10.1 A volume out of a stack of areas

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · BALLPARK

SCENE 32W

Rasmussen has the resurveyed surface areas at 3 levels 2 metres apart — 2.30, 2.45 and 2.60 square kilometres. Ferrand wants the volume held between the lowest and the highest of them.

GUIDE 74W

The numbers on offer are two ways of adding the three areas, a spacing and a conversion. Only one of the sums treats the two end areas the way the rule requires. Areas stack up through levels exactly as flows stack up through time, so the same weighting applies. The areas are the integrand here. An eleven per cent error in them is eleven per cent of headroom the control room thought it had.

ASK 13W

Estimate the volume held between 210 and 214 metres on the resurveyed curve.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the volume between two levels is the integral of the surface area with respect to level · the trapezoid rule adds half of each end value to all the interior ones · the definite integral as an accumulation, and its units — taken as read

VERDICT 88W

Volume is the integral of the surface area with respect to level, which is what a stage-storage curve is a table of. Stacking the areas with the trapezoid rule is the same arithmetic as stacking flows in time, with height in place of time. That is why the resurvey matters so much. The areas are the integrand. The 2003 sheet gives 2.58, 2.75 and 2.92 for the same 3 levels. It returns 11.0 million for this slice. That is 11% of headroom the control room believed it had.

TAKEAWAY 24W

A volume is an accumulation of areas up through the levels, so the area curve is the integrand and its errors are inherited whole.

10.2 The exponent is not the whole story

SPILLWAY & GATES · GATE HOUSE · A PERSON · PROPAGATE

SCENE 38W

The spillway rating is quoted to three figures and Wilkes cannot defend it. The head is gauged, the crest length is surveyed, the coefficient came out of a handbook, and there is one calibration slot in the week.

GUIDE 67W

The panel shows an error budget: each input's own width, times the power it is raised to in the rating, contributes a bar. One calibration slot is affordable this week, so read the bars rather than the inputs and buy the measurement that shrinks the tallest one. A term with a big exponent and a small width may be contributing less than a modest term measured badly.

ASK 16W

One calibration is affordable this week. Which of these is carrying the uncertainty in the discharge?

BEHIND THE BUTTON 161W

How the widths combine. In a product of powers, the fractional uncertainties add in quadrature, each multiplied by its exponent. Head enters at the power of three halves, so its fractional width counts one and a half times over. Crest length enters linearly. The coefficient enters linearly too. The exponent is a multiplier on the width, not a substitute for it.

Why the exponent alone misleads. Head has the largest exponent and is gauged continuously, so its width may be small. The handbook coefficient has an exponent of one and a width nobody has ever measured on this structure. Which dominates is an arithmetic question, and it is exactly the question the bars answer.

Why three figures was indefensible. A discharge quoted to three significant figures claims a fractional uncertainty under a per cent, and no combination of these inputs supports that. Finding which term is carrying the width is the first step to either measuring it or quoting fewer figures.

TAKES AS READ

a fractional uncertainty is carried into a product weighted by the power each term is raised to · a measured quantity and a handbook quantity are different kinds of number

VERDICT 98W

A factor contributes its exponent multiplied by its own fractional width. The head carries the three-halves that the whole spillway is famous for. It is gauged well enough that one and a half times 4% is still small beside one times 15%. Buying another head gauge therefore improves the term that was already good and barely moves the total. What makes this worth catching is that the shortcut — rank by exponent — is reasonable, teachable, and gives exactly the wrong answer here. The

> coefficient stays wide because nobody has ever calibrated this crest against a measured river.
>
> **TAKEAWAY** 29W
>
> A term contributes its own fractional width times its exponent. So a badly known factor with no power over it can outweigh a well-known one raised to three halves.

10.3 What the correction does not touch

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · CHOICE

SCENE 33W

The resurvey has moved every volume in the control room. Zawadzka is asked whether her uplift predictions move with them, and she has the foundation relation on the board beside the level gauge.

GUIDE 59W

The options differ in how far the correction travels: everything, one path, one measurement, or nothing. The test is not importance but dependency. Ask of each result whether the corrected quantity appears anywhere in the calculation that produced it. A correction made by feel either spreads to numbers it never touched, or stops before it reaches one it did.

ASK 10W

Which of these has to be recomputed after the resurvey?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

a correction propagates only to quantities computed through the thing corrected · the level is measured directly against a datum rather than derived from a volume

VERDICT 103W

The corrected quantity is the surface area. It enters exactly two calculations. The volume integrates it. The level forecast divides a net flow by it. Those move. The uplift relation takes the level as its input. The level is read against a datum on the wall rather than computed from a volume. So nothing in the gallery changes. The reservoir is at the same height it was yesterday and pushes with the same pressure. Tracing which results used the corrected term is the whole of the recovery. Doing it by feel is how a correction either spreads too far or stops too early.

> **TAKEAWAY** 22W
>
> A correction travels along the path the calculation took, so a quantity that never used the corrected term is untouched by it.

END OF THE DAY 22W

An integral inherits every error in its integrand, and the only quantities it spares are the ones that never went through it.

A day when nothing happens

PLAN CARD 117W

Thursday. For the first time in a fortnight, nothing has to be decided before lunch. Anand wants to raise the machines. From the rate alone she can say what that should buy her. That is a prediction, so it is a thing that can be wrong. Zawadzka has a routing expression that goes to zero over zero the instant a release starts. Zero over zero is not a small number. It is no number at all until it is handled properly. Ekundayo has a rating curve and a head rise of six centimetres. Today you predict with a tangent line, then measure what really happened. You also work one limit that no amount of substitution will reach.

BRIEFING 18W

A quiet day, which is the only kind on which a prediction can be checked against a measurement.

WHAT YOU DECIDE 17W

Predict with a tangent line and then measure it, and evaluate a limit that arithmetic cannot reach.

11.1 What the tangent line promised

POWER & DEMAND · POWERHOUSE · A ROOM · VERIFY

SCENE 42W

Anand is at 46 cubic metres a second through the machines and proposes 52. The slope of the output curve at 46 is 197 kilowatts for every extra cubic metre a second, and the tailrace traverse takes 2 hours of deck time.

GUIDE 58W

Lock a prediction for the gain before anything is changed: the panel will not let you look first, and nothing about the prediction can be edited afterwards. Then raise the flow, and then spend the two hours of deck time on the traverse that measures what actually happened. Not measuring is an answer too, and a poor one.

ASK 16W

Predict the gain from raising the flow, make the change, and report what it actually did.

BEHIND THE BUTTON 166W

Where the prediction comes from. The slope of the output curve at 46 cubic metres a second is 197 kilowatts per extra cubic metre a second. Multiplying that by the six-unit change is a tangent-line estimate, a linear approximation of a curve. It will be an overestimate or an underestimate, depending on which way the curve bends.

Why the tangent line is a promise with a range. A derivative is exact only at the point where it was taken. Six cubic metres a second is a long way along a curve that is bending. So the honest prediction carries a caveat about direction. The measurement is what tells you the size of the error.

Why the traverse costs and why that is the point. Two hours of deck time is real, and skipping it leaves a prediction standing as though it were a result. The format spends that cost deliberately: a prediction that is never checked cannot be wrong, which is precisely what makes it worthless.

TAKES AS READ

the tangent line at a point predicts nearby values of the quantity · a curve that bends away from its tangent makes that prediction worse the further it is carried

VERDICT 88W

A tangent line is the best straight-line description of a curve at one point. Over a small step it is excellent. Six cubic metres a second is not a small step here. The output curve is bending towards its own maximum, so every extra cubic metre buys less than the one before it. The straight line therefore runs above the truth for the whole interval. The error is not noise and does not average out. It is the second derivative, and it points the same way every time.

TAKEAWAY 16W

A tangent-line prediction is only checked by measuring, and its error is the curvature it ignored.

11.2 Nothing over nothing

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · DERIVE

SCENE 38W

Zawadzka has the arriving fraction of a step release written as 1 − e^(−t/T), with T 4 hours. The initial rate of arrival is that fraction divided by t, and at t = 0 both parts are 0.

GUIDE 56W

Build it a line at a time. Putting t = 0 into the expression gives nothing over nothing. So the value has to be got at as a limit rather than by substitution. Finish by saying what makes the method legitimate — the last step of this stop is naming the condition, not producing the number.

ASK 13W

Evaluate the initial rate of arrival, and say what makes the method legitimate.

BEHIND THE BUTTON 163W

What zero over zero means. It is not a value and it is not an error. It is a statement about the expression's value. That value depends on how the numerator and denominator approach zero relative to each other. Both go to zero here, and the ratio they approach is the thing being asked for.

Two legitimate routes. L'Hôpital's rule applies because both parts are differentiable and both tend to zero — differentiate top and bottom separately and take the limit again. Or expand the exponential as a series: 1 − e^(−t/T) is t/T minus smaller terms, so dividing by t leaves 1/T plus terms that vanish. Both give the same number and the series shows why.

Why naming the condition is the point. L'Hôpital's rule used without checking the indeterminate form is the most common way a correct-looking calculus answer turns out to be nonsense. A student who states the condition has the tool; one who applies it reflexively has a habit.

TAKES AS READ

a quotient whose numerator and denominator both go to zero has no value until it is worked · the derivative of e^(−t/T) with respect to t is −(1/T)·e^(−t/T) · limits, one-sided limits, and where a limit fails to exist — taken as read

VERDICT 88W

Zero over zero is a statement that substitution has failed, not a value. Differentiating the top and the bottom separately replaces both by the rates at which they are vanishing. The quotient settles on the ratio of those rates, which is why the answer depends on T at all. The result matters for the warning sheet. The arriving fraction is nearly nothing at the first instant. The rate at which it starts arriving is a definite quarter per hour that no reading at the start could ever show.

TAKEAWAY 29W

A limit in the form zero over zero is not zero and not one. Differentiating the top and the bottom separately is what gets a value out of it.

11.3 Six centimetres, in cubic metres

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A PERSON · BALLPARK

SCENE 31W

Ekundayo has the rating slope at 44.5 cubic metres a second for every metre of head. The gauge shows the head 6 centimetres higher than it was at the morning call.

GUIDE 77W

Five numbers, and two of them are the same rise written in different units. One is a slope, one is a head rather than a change in head, and one is a constant from the rating equation. A slope multiplied by a step only works if the step is in the units the slope was quoted in. Centimetres put into a per-metre slope give an answer a hundred times too large. It still looks like a discharge.

ASK 12W

Estimate the change in spillway discharge from a six-centimetre rise in head.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

a small change in a quantity is its rate of change multiplied by the small change in the input · 6 centimetres is 0.06 metres

VERDICT 88W

A differential is the simplest use a derivative has. Multiply the slope by the step and read off the change. It is exact only in the limit. Over six centimetres on a curve this gentle the error is under 1%, well inside what the gauge can resolve. The reason to write it as a slope times a step is that the slope is a number worth carrying. It says a centimetre is worth almost half a cubic metre a second today, and seven-tenths of one at three metres.

TAKEAWAY 28W

A differential is the tangent line's arithmetic — a slope times a step — and it is what turns a gauge reading into the quantity somebody cares about.

END OF THE DAY 28W

A derivative predicts nearby values, its own arithmetic is a multiplication, and a limit that comes out as zero over zero has to be worked rather than read.

DAY 12 OF 15

The front that never came

PLAN CARD 114W

Friday. The front that justified Monday's pre-release never arrived. The reservoir is short going into summer. Halloran has been polite about it in a way that is worse than not being. What is left is the recession. That is the river falling away with no rain behind it. One constant describes the whole fall. Ferrand has to solve the equation for it. Ekundayo has to fit that constant on days that have already happened. Then he has to live with it on days that have not, which is the harder half. Anand needs the answer as an hour, not as a curve. Today you solve the recession and turn its constant into a date.

BRIEFING 18W

The rain stopped on Wednesday and the river is falling on a curve with one number in it.

WHAT YOU DECIDE 14W

Solve the recession equation, fit its constant honestly, and turn it into a time.

12.1 The river falling away

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · DERIVE

SCENE 38W

Ferrand has the recession as $dQ/dt = -k \cdot Q$, with the flow falling in proportion to itself now that no rain is behind it. He wants the flow as a function of time before the weekend schedule is written.

GUIDE 54W

Build it a line at a time. The flow is falling in proportion to itself. That is the equation whose solution is an exponential. Separating the variables is what shows that rather than asserting it. Finish with the flow as a function of time, in a form the weekend schedule can be written from.

ASK 12W

Solve the recession equation for the flow as a function of time.

BEHIND THE BUTTON 164W

What 'in proportion to itself' means. With no rain behind it, the river is draining stored water, and the more there is the faster it leaves. $dQ/dt = -kQ$ says exactly that, and it is the same equation as radioactive decay, cooling coffee and a discharging capacitor. Recognising the form is most of the work.

Why the constant is a rate and not a time. k has units of one over time, so $1/k$ is the time the flow takes to fall to about a third of its value. Quoting the recession constant and quoting that characteristic time are two ways of saying one thing, and the second is the one an operator can feel.

What the schedule needs from it. Committing to a release for Saturday means predicting Saturday's natural flow. An exponential fitted now gives that with one number. The same function says how wrong you will be if the recession constant is off. That is the honest part of the promise.

TAKES AS READ

a differential equation with the variables separable can be integrated a side at a time · the antiderivative of one over a quantity is the natural logarithm of it

VERDICT 87W

Separating gives one over Q against a constant. The antiderivative of one over a quantity is the only case the power rule cannot do. Raising the exponent by one gives zero and the division fails, which is precisely the gap the logarithm fills. Exponentiating both sides then turns the constant from something added into something multiplied. That factor is the flow at the start. What comes out is the one curve on this site with no end to it, so operations quote k rather than a date.

TAKEAWAY 19W

A rate proportional to the quantity itself integrates to a logarithm, and exponentiating both sides turns that into decay.

12.2 The constant that fitted too well

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · HOLDOUT

SCENE 65W

A recession constant says how fast a river falls away once the rain has stopped. Ekundayo has a week of recession to fit one on, and four later days held back that the fit has never seen. The score is how well a constant reproduces the days it is scored against, so higher is better. Halloran wants the number that will be used next week.

GUIDE 72W

Drag the constant and watch how well it fits the week. Two things to hold on to. Each score comes from a handful of days. So a small difference between two constants is that week's weather rather than the river. And you are keeping the best of many constants. That is a contest luck can win. Then press freeze. Freezing commits you and opens the four held-back days. Report what those say.

ASK 16W

Choose the recession constant yourself, freeze it, and score it on days it has never seen.

BEHIND THE BUTTON 184W

What the constant describes. After the rain stops, a catchment drains what it has stored, and the flow falls away by roughly the same fraction each day. That fraction is the recession constant. It is a property of the soil, the geology and the shape of the valley. So it should be much the same next week as it was last week.

Why fitting a week fits the week. The fit sees one week of weather, and that week had its own showers, its own evaporation and its own gauge noise. A constant tuned to the last digit is tuned to those as well as to the catchment. The parts that came from the weather will not happen again.

What the freeze is for, and what would break without it. Held-back days can only test a constant that was chosen without them. If you could keep adjusting after seeing them, you would be fitting to those four days too, and there would be nothing left to test the choice on. That is the whole of the procedure: keep some data back, commit, then look.

TAKES AS READ

a constant chosen to fit a sample is chosen partly on that sample's noise · noise does not repeat, and a real feature of the river does · the definite integral as an accumulation, and its units — taken as read

VERDICT 95W

A constant chosen on a sample is chosen partly on that sample's noise, and noise does not come back. That is why the broad shoulder holds up and the narrow spike does not. The shoulder is a property of how this catchment drains, which next week also has. The spike is a property of seven particular days, which it does not. So a score measured where the constant was chosen is never a measurement of the constant alone, and it is wrong in the same direction every time. The cure is procedure rather than cleverness.

TAKEAWAY 24W

A score measured where the constant was chosen is a score of the constant plus the sample, and it is biased high every time.

12.3 How long until it halves

POWER & DEMAND · POWERHOUSE · A PERSON · BALLPARK

SCENE 35W

Anand has the recession constant at 0.22 a day and a maintenance window she can only take once the flow is below half of what it is now. She wants a date, not a curve.

GUIDE 73W

Five numbers and only two of them survive the rearrangement. The others are the factor the flow falls by, the fraction left, and the reciprocal of the constant. All are true; none is what the solved form asks for. Take the logarithm on paper first and see which quantities are left standing. The starting flow cancels, which is why the answer says the same thing about a big river and a small one.

ASK 10W

Estimate how long the recession takes to halve the flow.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the recession solution can be rearranged for the time at a given fraction of the start · the natural logarithm of 2 is about 0.693

VERDICT 94W

Setting Q equal to half of Q_0 and taking logarithms cancels the starting flow entirely, so the halving time depends on the constant alone. That independence is the whole reason a recession is quoted as a constant. It says the same thing about a river at 400 cubic metres a second and at 40. No absolute rate of fall does that. It also gives the shape of the wait. The next halving takes exactly as long as this one. So the machines wait three days for the first and three more for the second.

TAKEAWAY 21W

Rearranging an exponential solution for time replaces the curve with a single number that does not depend on where you started.

12.4 What the round is looking at now

SPILLWAY & GATES · GATE HOUSE · A ROOM · SPOT

SCENE 41W

The gate gallery inspection has more to look at than one round covers. The duty engineer sets the priority. It changes as the day does — after the level rises, after the trash rack blocks, after the outage window is confirmed.

GUIDE 54W

Inspect what the current priority wants and leave the rest for the next round. The priority in the gallery log changes during the day and nobody announces it. What is scored is the items either side of a change, because they are the only ones that show whether the round is reading the log.

ASK 12W

Inspect to the priority in the log, and keep watching the log.

BEHIND THE BUTTON 114W

Why the priority changes. On a normal day the round is about wear, because wear is what a scheduled inspection is for. When the level rises it is about anything that has to move under load. When an outage is confirmed it is about anything that can only be looked at with the water off. That is a list that is unavailable most of the year.

Why the changeover costs more here than most places. An outage window is measured in hours and comes round twice a year. An item not looked at during one is not looked at for six months, and nothing in the log says it was displaced rather than checked.

TAKES AS READ

some things can only be inspected when the plant is shut down

VERDICT 124W

Three priorities run across the day and each wants a different part of the gallery. Anything showing wear, then anything that has to move under load, then anything that can only be reached with the water off. An item can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change. Most of the gallery is wanted by neither priority. It is correctly left for next month by somebody who has read nothing. And this is the case where the cost is largest. An outage window is a few hours twice a year. An item skipped during one is unavailable until the next. The round sheet records only that the round was completed.

TAKEAWAY 17W

The cost of a withdrawn instruction is highest where the opportunity does not come round again soon.

END OF THE DAY 27W

A rate proportional to the quantity itself decays exponentially, and a constant fitted on one stretch of data is only honest when it is scored on another.

Three things before nine

BEFORE THE DAY — CANVASS

Who downstream actually got the siren test

The warning system was tested on Sunday and the emergency plan assumes the valley heard it. Ask along the downstream settlements until enough answers have come back to say whether the plan rests on a system that works or on a paragraph in a folder.

PLAN CARD 120W

Saturday. Three jobs have to close before the nine o'clock call. Baptiste has the inflow and the release drawn on one pair of axes. The two lines cross in the middle of the afternoon. Her committee will read the shaded region between them as though it were a total. Wilkes has a window between eight and two, when the inflow ran clear of the release. He wants to know what that window added. Ekundayo has the suspended load record from the peak. It is a fraction with the variable in both halves, and it will not integrate as written. Today you integrate a difference and read what its sign is telling you. One of the three needs a substitution first.

BRIEFING 16W

Two curves crossing at four o'clock, and a load record that will not integrate as written.

WHAT YOU DECIDE 19W

Integrate a difference, read its sign, and substitute your way through an integrand nobody can antidifferentiate as it stands.

13.1 The region and the number

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · CHOICE

SCENE 33W

Baptiste has inflow above release until four o'clock and below it afterwards, with the region between them shaded on the committee sheet. She wants the caption right before the sheet leaves the room.

GUIDE 65W

Three of the options describe the shaded region and one describes the integral, and today those are not the same object. The inflow runs above the release for part of the day and below it for the rest. Decide whether the quantity in question keeps its sign or throws it away. A caption that swaps the two overstates the day by exactly twice the afternoon.

ASK 13W

Across the whole day, what does the integral of inflow minus release give?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

the storage change over an interval is the integral of inflow minus release · an integrand that changes sign contributes negatively where it is negative · the definite integral as an accumulation, and its units — taken as read

VERDICT 82W

Where the inflow runs above the release, the integrand is positive and the reservoir fills. After four o'clock it is negative and the reservoir empties, and the integral subtracts that part. The shaded region on the sheet is the integral of the absolute difference. That is a larger number and not a storage change at all. It is the total water that moved either way. A caption calling the shading what the reservoir gained overstates the day by exactly twice the afternoon.

TAKEAWAY 21W

An integral of a difference is a signed total, so the afternoon subtracts from the morning rather than adding to it.

13.2 What the gap between them added

SPILLWAY & GATES · GATE HOUSE · A PERSON · BALLPARK

SCENE 33W

Wilkes has 6 hours from 8 until 2 where the inflow averaged 96 cubic metres a second and the release held at 62. He wants what the reservoir took on in that window.

GUIDE 72W

Five numbers, two of which are rates that belong on opposite sides of a subtraction, and one of which is those two added together. A window of gain is what one rate exceeds the other by, held for a length of time. Sort the numbers into rates and durations before doing anything with them. Adding the rates instead of differencing them describes a reservoir with no machines running and no gates open.

ASK 10W

Estimate the volume the reservoir gained between eight and two.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

the storage gained over an interval is the integral of the difference of two rates · 6 hours is 21,600 seconds · the definite integral as an accumulation, and its units — taken as read

VERDICT 90W

Two curves and one integral. The storage gained is the area between them. That is the integral of the difference rather than the difference of two integrals worked separately. It is the same number, and much less arithmetic. On a window where both rates hold steady the integral collapses to a rectangle, and the only thing that can go wrong is integrating the inflow alone. That gives 2.07 million and describes a reservoir with no machines running and no gates open, which is not the one anybody is standing in.

TAKEAWAY 23W

The volume between two rate curves is the integral of their difference, which for two steady rates is a rectangle and nothing more.

13.3 The integrand that had to be changed

CATCHMENT & INFLOW · CATCHMENT & INFLOW DESK · A ROOM · DERIVE

SCENE 34W

Ekundayo has the suspended load at the gauge as $L(t) = 60t \div (t^2 + 4)$ tonnes an hour, with t measured from the peak. Rasmussen wants the four-hour total for the resurvey file.

GUIDE 65W

You build the working a line at a time, choosing at each step the expression the line above actually gives you. Every wrong branch is legal algebra, so you cannot spot the answer by looking for a malformed line. The total load over four hours is an integral of the rate. The integrand here is the kind that wants a substitution rather than a rule.

ASK 16W

Evaluate the total suspended load carried past the gauge in the four hours after the peak.

BEHIND THE BUTTON 160W

Why a total is an integral. The gauge reports a rate — tonnes an hour — and the file wants tonnes. Multiplying a rate by four hours would be right only if the rate held still, and this one falls away from the peak. Integrating is what adding up a changing rate means.

Why substitution is the move. The numerator is a constant times t and the denominator is $t^2 + 4$, whose derivative is $2t$. That is the shape that says one variable will do. Let u be the denominator, and the whole integrand collapses to du over u. The integral of that is a logarithm.

What the answer is for. A resurvey file needs the mass that went past, because that is what has left the catchment and arrived somewhere downstream. It is the number that will be compared against how much the reservoir actually silted up. Being able to defend the working matters more than the figure.

TAKES AS READ

an integral can be rewritten in a new variable if the differential is converted with it · the antiderivative of one over a quantity is the natural logarithm of it

VERDICT 94W

The integrand resists because it is a quotient with the variable in both parts. It stops resisting the moment somebody notices the top is very nearly the derivative of the bottom. Substituting turns it into one over u, whose antiderivative is a logarithm. The whole difficulty was a change of variable rather than a new technique. What has to travel with the variable is the differential and both limits. It is the limits that get left behind, and an integral evaluated at the old ends of a new variable is wrong without looking wrong.

TAKEAWAY 19W

> A substitution replaces the variable and its differential together, and the limits of integration have to travel with them.

END OF THE DAY 23W

An integral of a difference carries the difference's sign, and an integrand that resists is usually one substitution away from an ordinary one.

DAY 14 OF 15

The last reversible moment

PLAN CARD 112W

Sunday night. The last decisions that can still be unmade are made now. Berg wants the night orders written against the rate the level is climbing. She does not want them written against the level itself. A level threshold gives no warning. A rate does. Ferrand has the net flow as a formula and needs the hour when the reservoir is fullest. In the gate chamber the only trace on the screen is the net flow. The level plot has not been drawn since Friday. Wilkes has to say what the level did from that rate alone. Today you write rules on a rate, and find the hour an accumulation tops out.

BRIEFING 19W

Everything on the board tonight is a derivative, and the decisions have to be written before the data arrives.

WHAT YOU DECIDE 21W

Write rules on a rate, find where an accumulation is greatest, and read a level off a graph of its derivative.

14.1 Rules on the rate, not the level

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A ROOM · TRIGGER

SCENE 33W

The night board is blank. Berg wants the farm clearance set against how fast the level is climbing rather than against how high it is, and signed before the eight o'clock update arrives.

GUIDE 83W

One rule, written on the rate the level is climbing rather than on the level itself, and fixed once the board is released. Moving stock and machinery off the flood plain takes five days from the moment it is called. Only the early updates arrive with five days left. So the line has to be low enough that one of them crosses it. It also has to be high enough not to have been crossed on a night the river was doing nothing.

ASK 18W

At what rate of rise do you call the farm clearance, given it needs five days to happen?

BEHIND THE BUTTON 159W

Why the rate rather than the level. A level threshold fires when the water is already there. A rate threshold fires while it is on its way. The earlier warning is worth more than the certainty. A reservoir climbing fast at a moderate level is a different situation from a still one near the top. A level rule cannot tell them apart.

What lead time means on a board like this. Every action has a duration — notifying downstream, opening gates in sequence, clearing the tailrace path. A stage set too close to the limit begins an action that cannot finish, which is worse than no rule because it is a rule somebody trusted.

Why release before the updates. A threshold adjusted while the data comes in is not a threshold; it is a commentary on decisions already taken. Signing the board first is what makes the night's actions attributable to a rule rather than to whoever was awake.

TAKES AS READ

a threshold on a rate fires earlier than a threshold on the quantity it is filling · a staged action has a lead time that cannot be compressed once it has started · the definite integral as an accumulation, and its units — taken as read

VERDICT 95W

Two ways of getting this wrong look nothing alike and cost the same. A threshold set high never fires and the stage simply does not happen. A threshold set at a number that sounds serious fires on Thursday with 72 hours left. That is a correct rule for an action that needed 120. So the rule is satisfied and the clearance is already impossible. Writing the board on the rate rather than on the level is what makes any of it reachable. The rate crosses its threshold two days before the level crosses its own.

TAKEAWAY 28W

A rule written on a derivative fires while there is still time to act; the same rule written on the level fires when the level is already there.

14.2 Where the accumulation tops out

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · DERIVE

SCENE 39W

Ferrand has the net flow as $N(t) = 40 - 8t$ cubic metres a second, with t in days from tonight. The storage is everything that net flow has added since. He wants the day the reservoir is fullest.

GUIDE 62W

Build it a line at a time. The storage is the accumulation of a net flow that is falling. So the storage is greatest when the net flow stops being positive. That has to come out of the working rather than out of a guess. Then classify the turning point, because 'fullest' is a maximum and the algebra has to say so.

ASK 15W

Find the day the accumulated storage is greatest, and establish that it is a maximum.

BEHIND THE BUTTON 164W

How the two are related. If S is what N has added since tonight, then N is the rate at which S is changing. That is the fundamental theorem, in the direction people use least. So questions about S's maximum are questions about where N crosses zero, and no integration is needed to find it.

Why the sign matters more than the value. While the net flow is positive the reservoir is filling; once it is negative it is emptying. The day the fullest level occurs is the day the sign changes. That is why a linear net flow makes this a one-line answer. A real one makes it a root-finding problem.

What it is worth. A dam operator plans drawdown around the day storage peaks, because that is the last moment a decision can be made with the season's water still in hand. Being a day out on it is a real cost, and it is why the classification is asked for explicitly.

TAKES AS READ

an accumulation function has the rate that built it as its own derivative · a quantity turns over where its derivative crosses zero

VERDICT 90W

The fundamental theorem does the whole of the first step. The derivative of an accumulation with a moving upper end is the rate at that end. So nothing has to be integrated to find where the total turns over. That is the practical value of it here. The reservoir's fullest hour is a property of the net flow alone, readable from a gauge. The storage figure itself is never needed. The second derivative is constant and negative, so there is one turning point and it can only be a top.

TAKEAWAY 28W

The peak of an accumulation is where the rate that fills it crosses zero, so the whole question is answered from the rate without ever evaluating the total.

14.3 The only trace on the screen

SPILLWAY & GATES · GATE HOUSE · A PERSON · CHOICE

SCENE 36W

The level plot has not been drawn since Friday. What Wilkes has is the net-flow trace, and a handover to write. The trace is positive and falling all morning, crossing zero at noon, negative all afternoon.

GUIDE 65W

Every option is a shape the level might have taken, and each fits some part of the trace. What separates them is one rule: the sign of the net flow gives direction, and its size gives steepness. A falling positive trace is still a rising reservoir. The early morning is where this goes wrong most easily. The largest net flow sits furthest from the peak.

ASK 10W

From the net-flow trace alone, what did the level do?

BEHIND THE BUTTON 198W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. That was 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason is indistinguishable from a sound one. That stays true until the day the approximation is what is being tested.

TAKES AS READ

the level rises where the net flow is positive and falls where it is negative · a quantity turns over where its rate crosses zero, not where its rate is largest

VERDICT 88W

Positive net flow means the level is climbing, whatever the net flow is doing itself. So a falling positive trace is still a rising reservoir, climbing less eagerly. That is a statement about curvature and not about direction. The level turns over where the trace crosses zero and nowhere else. The trap is the early morning, where the net flow is at its largest. That is where the level is climbing fastest. It is the steepest part of the level curve and the furthest thing from its peak.

TAKEAWAY 24W

> A rate's graph carries the whole shape of what it built, and the rate's own peak is nowhere near the peak of the quantity.

END OF THE DAY 26W

A rate written into a rule buys the lead time a level cannot, and an accumulation's whole shape is readable from the rate that built it.

What the fortnight is worth

PLAN CARD 99W

Monday, a fortnight on. The reservoir is at 91%. The reach was warned twice and flooded neither time. The resurvey has moved every volume on the site by 11%. Zawadzka has three fits of the uplift against head. The one with the best summary number is not obviously the one to keep. Ferrand has to reduce 14 days to a single figure for the annual return. And the liaison committee sits this evening. Today you judge a fit by what it leaves over. Then you reduce a fortnight to one rate, and say what these two weeks actually settled.

BRIEFING 15W

The water has gone and what is left is what can be defended in writing.

WHAT YOU DECIDE 20W

Judge a fit by what it leaves over, reduce a fortnight to one rate, and say what has been established.

15.1 A pattern is not a scatter

STRUCTURE & SEEPAGE · SEEPAGE & UPLIFT BAY · A ROOM · RESIDUAL

SCENE 37W

Three fits of the uplift against reservoir head are on Zawadzka's screen, within a tenth of a kilopascal of each other on the summary statistic. The residuals are plotted where each gauge actually sits under the wall.

GUIDE 65W

Three fits, and their summary statistics sit within a tenth of a kilopascal of each other. The number underneath cannot choose between them. What can is the residual field. Each stick is one uplift gauge, drawn where it sits under the wall. Up for positive and down for negative. Look at all three fields, then carry the fit whose leftovers look like nothing in particular.

ASK 16W

Look at each field before choosing. Which of these fits would you carry into next season?

TAKES AS READ

a complete model leaves residuals that look like noise · a fitted parameter can absorb a systematic effect without removing it

VERDICT 89W

Two of these three leave residuals that are positive on one side of the wall and negative on the other. That is not scatter. It is a quantity the model does not contain. Adding a free parameter that soaks it up improves the summary number. It leaves the cause exactly where it was. It will then bias every extrapolation in the same direction, quietly, because nothing about the fit looks wrong afterwards. One large residual is a bad gauge; six small ones in a pattern are a missing term.

TAKEAWAY 25W

A low summary statistic with a pattern under it hides a term the model is missing, and it will bias every prediction the same way.

15.2 14 days, one number

STORAGE & LEVEL · STORAGE & LEVEL BOARD · A ROOM · BALLPARK

SCENE 33W

Ferrand has 47.2 million cubic metres through the site over 14 days and 1 line on the annual return for it. He wants the steady flow that would have delivered the same water.

GUIDE 75W

The numbers offered are one volume, two ways of writing the interval, and a peak that belongs to a different question. An average value is an accumulation over a length, and the length has to be in the units the flow is quoted in. Days put into a per-second figure are out by a factor of 86,400. A peak compares two instants; an average compares two events, which is what the return is asking for.

ASK 12W

Estimate the average inflow across the fortnight, in cubic metres a second.

BEHIND THE BUTTON 109W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before it is placed. It is also how a quantity belonging to a different part of the problem is caught. That is the habit this format exists to build.

TAKES AS READ

an average value is an accumulation divided by the length of its interval · 14 days is 1,209,600 seconds · the definite integral as an accumulation, and its units — taken as read

VERDICT 87W

This is the day-eight arithmetic on a longer interval. Nothing about it changes with the scale. An average value is an accumulation over a length, whether that length is six hours or a fortnight. It is also the honest way to compare this flood with the record, because a peak compares two instants and an average compares two events. 40 is a wet fortnight here and not a remarkable 1. That is the sentence the return needs, and the peak of 136 would never have supplied it.

TAKEAWAY 23W

The same operation that reduced a morning to one flow reduces a fortnight to one, and neither is the mean of the readings.

15.3 What is worth writing down

DOWNSTREAM & PUBLIC SAFETY · DOWNSTREAM WARNING DESK · A PERSON · SCIENCETANK

SCENE 37W

The liaison committee sits at seven. Halloran has a fortnight of decisions to account for. Berg has a release she asked for on the first day. The authority will fund one piece of work out of it.

GUIDE 114W

The provisional correction to the stage-storage curve is 11 per cent at working levels and rests on 11 transects. A full resurvey is four weeks of boat time. Every volume and every level forecast on this site divides by that curve; the uplift and the seepage do not. The spillway coefficient is still a handbook figure carrying 15 per cent. It is the widest term in the discharge and the one nobody has measured here. The upper catchment has two gauges, and the arrival times they fed were checked against the reach for the first time in nine years. The uplift relation has a fortnight of high-level data it has never been fitted on.

ASK 13W

Spend what the authority will release on what the fortnight showed was missing.

BEHIND THE BUTTON 194W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely. It says what is worth hedging against. It says what is not worth pursuing at all. The last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a measurement that every other quantity is divided by is worth more than a measurement of one quantity

VERDICT 110W

The two worth funding are the two that everything else is computed through. The area curve is the integrand of every volume and the divisor of every level forecast. So an 11% error in it is an 11% error in all of them at once. And 11 transects is a provisional number the room is already relying on. The coefficient is the widest term in the discharge by a long way, and improving the head gauge instead would refine

> the term that was already good. More gauges would improve a forecast that was not what failed, and the uplift re-fit is real work that can wait for a quiet quarter.
>
> **TAKEAWAY** 26W
>
> What a fortnight establishes is whatever was measured well enough to be used again, and that is usually not the thing that was most argued about.

END OF THE DAY 19W

A model is judged by the structure it leaves behind, not by the size of what it leaves behind.

THE CLOSING CARD

The rain stopped on the ninth day with the reservoir at 91%, and the reach below the dam was warned twice and flooded neither time. The resurvey moved every volume on the site by 11%, so the storage curve the night orders are written against is the real one now. Ashfell keeps its gates, its gauges and one more season of margin than it had a fortnight ago.

What it cost: two villages packed and unpacked for a warning that turned out to be right to give, nine days of a crew on twelve-hour watches, and a spillway relation still fitted to data taken below half gate. What is unfinished: the uplift fits disagree above 90%, the 2003 survey is still what every drawing in the building shows, and the rule that would have caught all of it — order against the rate, not the level — is a line in a handover book until the next duty engineer makes it a habit.

And it held because of you. You worked the rate rather than the level, you fixed the volumes on a survey rather than on a drawing from 2003, and you ordered gates six hours ahead of a river that rises in two. Four villages went to bed dry through nine days of rain. That was you, every morning of it.

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